

GALVESTON, TX, USA

WMO: 722420

Lat: 29.273N

Lon: 94.859W

Elev: 5

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.5	39.6	23.0	17.4	42.3	27.6	21.5	46.7	29.0	47.8	26.9	47.0	15.5	20	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	92.1	79.4	90.9	79.5	90.1	79.4	82.2	88.2	81.5	87.4	81.0	86.9	11.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.6	84.8	79.7	154.4	84.8	79.2	151.6	84.8	45.7	88.6	45.0	87.8	44.5	86.9	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	23.1	20.2	31.6	96.0	3.8	3.0	28.9	98.2	26.7	100.0	24.5	101.7	21.8	103.9
			28.3	83.2	3.4	0.8	25.8	83.8	23.8	84.3	21.8	84.8	19.3	85.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	55.5	58.6	64.6	70.9	77.6	83.3	84.7	85.4	82.1	75.0	65.2	57.8
	DBStd	12.20	8.26	7.88	6.65	5.18	3.79	2.65	2.85	2.52	3.71	6.45	7.89	8.23
	HDD50	96	44	20	4	0	0	0	0	0	0	0	4	24
	HDD65	957	305	198	87	15	0	0	1	0	0	9	93	249
	CDD50	8054	214	262	458	627	857	998	1075	1098	963	775	462	265
	CDD65	3438	9	20	76	192	392	548	611	633	513	319	100	25
	CDH74	34771	3	11	87	619	3045	6450	7823	8202	5723	2507	279	22
	CDH80	12954	0	0	3	43	547	2432	3458	3856	2058	544	13	0
Wind	WSAvg	11.2	11.7	11.8	12.0	12.7	12.5	11.2	9.8	9.5	9.8	11.0	11.1	11.5
Precipitation	PrecAvg	43.4	3.4	2.3	2.6	2.4	3.2	4.2	3.5	4.6	6.1	3.7	3.4	3.7
	PrecMax	64.3	10.7	8.3	10.7	10.4	11.0	14.8	17.5	26.7	24.3	11.8	14.3	11.1
	PrecMin	19.9	0.2	0.0	0.1	0.0	0.0	0.2	0.0	0.1	0.3	0.0	0.4	0.5
	PrecStd	11.3	2.4	1.8	2.3	2.2	2.7	3.5	3.1	4.4	4.4	3.0	2.6	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.9	75.7	78.9	83.5	88.0	92.6	92.5	94.4	93.2	88.5	81.6	76.7
		MCWB	67.0	68.9	70.3	71.9	76.5	78.1	79.2	79.2	78.1	78.4	74.5	70.9
	2%	DB	71.0	72.8	76.2	81.1	85.8	90.4	91.2	92.5	90.4	86.6	79.4	73.4
		MCWB	65.8	68.4	70.3	72.7	76.5	78.7	79.6	79.8	78.8	77.8	73.1	69.6
	5%	DB	68.5	70.5	74.5	79.2	84.3	89.1	90.3	91.3	89.1	84.5	77.2	71.8
		MCWB	64.9	66.6	69.2	72.3	76.1	78.8	79.7	79.9	78.7	76.7	72.2	68.6
	10%	DB	66.1	68.3	72.6	77.5	82.9	88.0	89.6	90.3	87.8	82.9	75.1	69.8
		MCWB	63.4	65.3	68.2	71.3	75.6	78.7	79.7	79.7	78.0	75.8	70.7	67.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	71.6	73.4	76.9	80.1	82.5	82.5	83.1	82.5	81.5	77.3	73.0
		MCDB	71.9	74.2	76.4	79.8	84.1	88.0	89.1	89.6	88.0	84.8	79.9	75.2
	2%	WB	67.5	69.9	72.0	75.3	78.7	81.7	81.6	82.2	81.4	80.0	75.0	71.2
		MCDB	69.5	71.9	74.6	78.8	82.9	86.7	88.2	88.8	85.9	83.8	77.5	72.9
	5%	WB	66.1	67.8	70.8	74.3	77.9	80.9	81.1	81.4	80.7	78.7	73.4	69.8
		MCDB	67.7	69.5	73.2	77.7	82.3	85.9	87.7	88.2	85.4	83.0	76.6	71.1
	10%	WB	64.2	66.0	69.6	73.3	77.1	80.1	80.5	80.7	79.9	77.6	71.9	67.4
		MCDB	65.4	67.6	71.9	76.5	81.6	85.4	87.0	87.6	85.0	82.2	74.7	68.9
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.2	10.6	10.2	9.2	8.8	9.2	9.6	9.7	10.7	11.5	11.8
		MCDBR	12.9	11.8	10.1	10.2	9.5	9.7	9.9	10.4	10.3	9.8	11.0	12.7
		MCWBR	10.3	9.0	6.8	6.0	4.8	4.4	4.2	4.3	4.7	6.0	7.4	10.5
	5% WB	MCDBR	12.0	10.8	9.1	8.6	8.1	8.4	8.8	9.5	9.0	9.5	10.5	12.3
		MCWBR	10.8	9.3	6.9	6.3	4.8	4.7	4.6	4.6	4.7	7.0	8.2	11.2
Clear-Sky Solar Irradiance	taub	0.351	0.366	0.380	0.434	0.475	0.465	0.487	0.472	0.450	0.387	0.375	0.359	
	taud	2.510	2.479	2.426	2.276	2.200	2.285	2.244	2.287	2.312	2.476	2.456	2.494	
	Ebn at Noon	281	286	288	274	262	263	257	260	263	275	269	272	
	Edh at Noon	29	32	36	43	47	43	44	42	40	32	30	28	
All-Sky Solar Radiation	RadAvg	924	1053	1449	1732	1961	2030	1925	1842	1589	1408	1068	824	
	RadStd	91	172	124	131	128	145	176	143	167	164	108	95	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	+212	+175

Nomenclature: See separate page